

BASE MONTALYA, Antarctica

WMO: 890560

Lat: 62.195S

Lon: 58.974W

Elev: 157

StdP: 14.61

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.0	4.2	-4.8	4.3	0.7	-0.3	5.4	4.9	58.0	14.9	52.5	14.1	25.8	100	1.221

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.8	39.0	37.8	37.8	36.9	37.3	36.5	38.1	38.5	37.4	37.7	36.7	37.0	21.2	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
37.7	33.4	38.1	37.1	32.6	37.5	36.2	31.5	36.7	14.6	38.7	14.2	37.8	13.9	37.1	43.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 50.2	45.1	40.4	DB	-2.1	43.8	5.7	3.7	-6.2	46.4	-9.5	48.6	-12.7	50.7	-16.9	53.4	(4)
(5)			WB	-2.4	40.9	5.6	1.7	-6.5	42.1	-9.7	43.1	-12.9	44.1	-16.9	45.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	34.1	34.3	32.7	29.1	27.1	23.3	21.4	22.4	23.8	27.1	29.8	32.1
	DBStd	6.82	1.85	2.32	3.42	5.48	5.14	7.00	7.99	7.39	6.79	4.35	2.74	2.22
	HDD50	8007	494	440	535	628	709	800	885	857	786	711	606	556
	HDD65	13482	959	860	1000	1078	1174	1250	1350	1322	1236	1176	1056	1021
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	19.9	16.6	17.5	18.9	20.5	19.7	21.8	20.6	21.8	22.5	21.7	19.4	17.6
Precipitation	PrecAvg	40.9	3.2	3.2	4.7	4.0	3.8	3.0	2.8	3.0	3.2	3.9	3.3	2.8
	PrecMax	54.8	4.5	4.9	7.4	6.0	6.3	4.5	4.3	5.6	5.4	6.7	4.5	4.5
	PrecMin	33.7	1.9	1.9	2.8	2.3	1.7	1.6	1.3	0.6	1.1	2.0	1.2	1.2
	PrecStd	4.6	0.6	0.7	1.1	0.9	1.1	0.7	0.9	1.1	0.9	1.0	0.8	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	40.3	40.6	37.7	36.0	34.2	33.5	33.6	34.0	34.3	36.0	39.2
		MCWB	38.9	38.9	38.4	36.7	34.9	33.3	32.8	32.6	33.3	33.4	34.7	37.7
	2%	DB	38.5	38.7	38.5	36.5	34.9	33.2	32.3	32.3	32.5	33.3	34.7	36.9
		MCWB	37.3	37.7	37.5	35.7	34.1	32.7	31.9	31.7	32.1	32.6	33.8	35.8
	5%	DB	37.5	37.8	37.6	35.7	34.0	32.3	31.7	31.6	32.0	32.4	33.8	35.8
		MCWB	36.5	36.9	36.9	35.0	33.5	31.9	31.3	31.1	31.7	31.9	33.2	34.7
	10%	DB	36.7	37.3	36.9	34.8	33.2	31.7	30.6	30.6	31.3	31.9	33.0	34.9
		MCWB	35.7	36.5	36.3	34.2	32.8	31.3	30.3	30.3	30.9	31.6	32.4	33.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	39.4	39.1	37.0	35.5	33.8	33.1	33.1	33.4	33.9	35.1	37.9
		MCDB	39.9	39.9	39.5	37.2	35.7	33.9	33.3	33.3	33.5	34.1	35.5	38.6
	2%	WB	37.7	38.0	37.8	36.0	34.3	32.8	32.2	32.0	32.4	32.8	34.1	36.0
		MCDB	38.1	38.3	38.0	36.2	34.5	33.0	32.3	32.2	32.5	32.9	34.3	36.3
	5%	WB	36.8	37.3	37.1	35.2	33.6	32.1	31.3	31.2	31.8	32.2	33.3	35.0
		MCDB	37.1	37.6	37.3	35.5	33.8	32.3	31.5	31.4	31.9	32.3	33.5	35.3
	10%	WB	36.0	36.7	36.3	34.3	32.8	31.3	30.4	30.4	31.1	31.6	32.5	34.2
		MCDB	36.4	36.9	36.5	34.7	33.0	31.5	30.6	30.6	31.2	31.7	32.7	34.5
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.8	4.4	5.0	5.8	6.8	7.6	7.2	6.8	4.7	3.7	3.4
		MCDBR	4.6	4.4	4.7	4.6	5.4	5.3	6.4	6.4	5.4	4.1	4.4	4.5
		MCWBR	4.0	4.1	4.4	4.5	5.3	5.3	6.7	6.7	5.5	4.1	4.1	3.9
	5% WB	MCDBR	4.3	4.1	4.5	4.3	5.3	5.0	6.5	6.2	5.1	4.0	4.1	4.4
		MCWBR	4.0	4.1	4.6	4.5	5.4	5.2	6.9	6.5	5.4	4.1	4.0	4.0
Clear-Sky Solar Irradiance	taub	0.307	0.306	0.296	0.279	0.224	0.365	0.210	0.267	0.286	0.308	0.302	0.308	
	taud	2.497	2.505	2.505	2.377	2.101	2.828	2.050	2.243	2.374	2.405	2.462	2.470	
	Ebn at Noon	307	292	268	215	141	96	140	210	264	287	307	311	
	Edh at Noon	31	28	23	16	10	6	10	18	25	30	32	33	
All-Sky Solar Radiation	RadAvg	1590	1147	617	270	78	33	56	201	529	981	1503	1801	
	RadStd	146	88	51	22	7	1	3	17	42	65	91	144	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.41	-0.70	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	-0.47	-0.76	N/A	N/A	N/A	N/A

Nomenclature: See separate page